

SEEDLINE · REQUIREMENTS

SeedLine requirements

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DOCUMENT CONTROL

Document ID	SDL-REQ-001	Version	0.7
Status	Draft	Date	2026-10-02
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/seedline		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.7	2026-10-02	Amish Chadha	Figures from SDL-CAL-001 v0.5 after the decisions of 2026-10-02 (13 mm slot with liners, UV-stabilized door): R11, R16 and R17 values updated; no status changed
0.6	2026-10-02	Amish Chadha	R17: USD 219.50 accepted for the prototype (SDL-DDR-003 A3, 2026-10-02); no status changed
0.5	2026-10-01	Amish Chadha	Status from SDL-CAL-001 v0.3 on the constructable design (SDL-DDR-003); R16 and R17 reported against value-engineering targets
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	Apply SDL-DDR-001 (R3 kept with skip-cell plates, R5 redefined for pelleted small seed, R6 materials, R13 marker arm, R18 bolted frame); status from SDL-CAL-001
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with concept status against each

0.1	2026-09-24	Amish Chadha	Initial scaffold
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SeedLine requirements

These requirements are proposals checked on paper at TRL 3 in SDL-CAL-001. They are not yet validated with users and will be revised after co-design sessions (see SDL-PRB-001). Version 0.3 applies the decisions Amish made on 2026-09-25 (SDL-DDR-001): the first user group is smallholder field crops, R3 is kept and met with skip-cell plates, and R5 is redefined so that seed under 2 mm is sown pelleted. Version 0.4 applies the recommendations Amish accepted on 2026-09-25 (SDL-DDR-002): the design-case walking speed becomes the 2.9 km/h walking-speed rule (R1), R8 is restated for ploughed field-crop seedbeds, R3 relies on the optional ratio kit for its full range, and the status column follows SDL-CAL-001 v0.2. Version 0.5 takes the status from SDL-CAL-001 v0.3, re-run on the constructable design (SDL-DDR-003); no requirement text changes, and R16 and R17 are reported against their value-engineering targets (STANDARDS section 18).

The **design case** is maize at 250 mm in-row spacing on 0.75 m rows, sown 50 mm deep in a tilled loam seedbed at a walking speed of 2.9 km/h (about 0.81 m/s). **Walking-speed rule** (SDL-DDR-002 item 5): walk at about 2.9 km/h or slower with the 0.8 and 1.0 ratios, and about 2.4 km/h or slower with the 1.2 ratio, so the cells stay within the assumed 0.30 m/s fill limit. The **small-seed case** is pelleted carrot or lettuce seed (about 3.3 mm pellets) at 25 to 50 mm spacing on a prepared vegetable bed; vegetable plates are the second plate set.

■ Table 1. Requirements

Status is from SDL-CAL-001 v0.3: met, at risk, not met, or not verifiable at TRL 3; the cost requirements are reported as over or under their value-engineering target.

ID	Requirement	Target	Verification	Status (SDL-CAL-001)
R1	Single-seed placement, design case	Misses 5 % or less and multiples 5 % or less of cells, in the terms of ISO 7256-1	Cell-speed check (CAL section 3); bench test on a sticky belt at TRL 4	At risk: cell speed 0.299 m/s at the 2.9 km/h walking-speed rule against an assumed 0.30 m/s limit, no margin; misses and multiples cannot be calculated
R2	Spacing uniformity	Coefficient of variation of in-row spacing 30 % or less in the design case	Monte Carlo (CAL section 10); field count later	At risk: 31 % of all spacings with the assumed miss and double rates; 9 % for single spacings
R3	Spacing range	25 to 400 mm in-row spacing, set by plate cell count (including skip-cell plates of one or two cells) and, where needed, one sprocket change from the optional ratio kit (SDL-DDR-002 item 4)	Spacing formula (CAL section 2)	Met with the ratio kit: 22 to 589 mm nominal, every target from 25 to 400 mm within 11.1 %; the base seeder (15 T only) covers 26 to 471 mm, within 17.3 %
R4	Spacing follows travel	Seed spacing set by distance travelled, not speed; wheel slip 8 % or less	Traction estimate (CAL section 6); field test later	Met: skid 1.2 to 1.7 %; rolling radius uncertain by up to 6.7 %, so calibrate per seedbed
R5	Seed size range	Raw seed from 3.5 to 15 mm in its largest dimension with plates of the same diameter; seed under 2 mm sown as pelleted seed (redefined, SDL-DDR-001 item 6)	Cell tolerance rule (CAL section 4); bench test later	Met on paper: 3.5 to 15 mm raw seed; pelleted carrot fits; raw 1 to 2 mm seed does not
R6	Printable plates	Each plate prints in 2 h or less on a 180 x 180 mm bed with no supports, in PETG (default) or ASA (strong sun); PLA for trials only; cell size within ± 0.2 mm	Print estimate (CAL section 4); measurement later	Met: 1.46 h (maize) to 1.96 h (groundnut), no margin for thick plates; accuracy not verifiable at TRL 3
R7	Crop change	Plate swapped in 2 min or less by hand, with no tools, without emptying more than the seed in the housing	Design review; timed trial later	Not verifiable at TRL 3: side door and hand knob are in the model; one 13 mm slot takes the 6, 8 and 10 mm plates with a printed liner pair
R8	Sowing depth	Adjustable 10 to 60 mm in 10 mm steps, held within ± 10 mm on a ploughed field-crop seedbed (restated, SDL-DDR-002 item 6; rough, hand-tilled seedbeds to be reviewed with partner data)	Depth geometry (CAL section 9); field check later	Met on paper: ± 10 mm holds for wheel-path bumps within about ± 15 mm under the drive wheel, taken as the ploughed-seedbed condition
R9	Work rate	0.1 ha/h or more in the design case, including turns and refills	Field capacity (CAL section 8)	Met: 0.142 ha/h (7.1 h/ha) at 2.9 km/h
R10	Push effort	Horizontal push force 150 N or less in the design case	Planar statics with soil model (CAL section 6); spring-scale test later	At risk: 132 N along the handle in the design case; 178 N at best on a heavy, loose seedbed

ID	Requirement	Target	Verification	Status (SDL-CAL-001)
R11	Handling	Mass 15 kg or less; handle grip height adjustable from 850 to 1,050 mm; lifted and turned at a row end by one person	Mass build-up (CAL section 5); weighing later	Met: 14.1 kg base, 14.9 kg with the marker kit (0.12 kg margin) with the 22 x 1.2 mm handle (SDL-DDR-002 item 3), weighed from the constructable model
R12	Hopper	2 L or more, fills from a jug without spilling, empties in 1 min or less when changing crops	Model volume (CAL section 8)	Met: 2.42 L
R13	Row-spacing kit	Telescoping marker arm (decided, SDL-DDR-001 item 3) sets the next row at 200 to 900 mm from the current row, within ± 25 mm	Design review	Met on paper
R14	Guarding	Chain and sprocket nip points covered on the side facing the operator's hands and feet; no exposed sharp edges above the opener	Design review	Met on paper: printed shroud with an outboard face plate over both sprockets, the chain and the tensioner's full travel
R15	Durability	Frame, drive and opener last 5 seasons of about 2 ha each with only chain, brush and plate replacement; plates last 1 season or more	Stress checks (CAL section 7); wear test later	Not verifiable at TRL 3: stresses are low, wear life unknown
R16	Prototype cost	Prototype parts against the USD 400 value-engineering target (<code>budget_usd</code>)	Priced BOM (<code>bom/</code> <code>bom.csv</code>)	USD 266.10 estimated, USD 133.90 under the target
R17	Replication cost	Base seeder (all items except the row marker kit and the optional ratio kit) about USD 200 in parts, a value-engineering target	Priced BOM	USD 224.10 estimated, USD 24.10 over the target; USD 219.50 accepted for the prototype on 2026-10-02 (SDL-DDR-003 A3), and the plate liners and UV-stabilized door added USD 4.60 after it, with the bronze bushing and go-kart kit savings to try at TRL 4
R18	Local build	Frame, handle and opener made from common steel sections with a drill and bolts (default, SDL-DDR-001 item 8) or a welder; drive from standard #35 roller chain or bicycle parts	Design review	Met on paper

■ Requirements not met or at risk

No requirement is not met on paper after SDL-DDR-003. Three are at risk, one is met with a thin margin and one cost requirement is over its value-engineering target:

- **R1 at risk:** under the walking-speed rule (2.9 km/h) the maize cells move at 0.299 m/s, at the assumed 0.30 m/s fill limit with no margin, and the limit itself needs a bench test. Plate size is revisited only with bench data.

- **R2 at risk:** the all-spacings CV depends on miss and double rates that need a bench test.
- **R10 at risk:** the design case is inside 150 N with a 12 % margin on assumed soil values; a heavy, loose seedbed needs about 178 N.
- **R11 thin margin:** 14.9 kg with the marker kit against 15 kg (0.12 kg margin), with the lighter 22 x 1.2 mm handle.
- **R17 over its value-engineering target:** base parts USD 224.10 against about USD 200, USD 24.10 over. The bearings, hubbed sprockets, tensioner, clips and fixings that make the design buildable added USD 22.00 (SDL-CAL-001 section 11); savings worth trying are in the design decisions register (SDL-DEC-001). On 2026-10-02 Amish accepted USD 219.50 for the prototype (the liners and UV-stabilized door then added USD 4.60); the bronze bushing and go-kart kit savings are tried at TRL 4.
- **R8 restated:** met on paper for ploughed field-crop seedbeds; the rigid opener follows the wheels, so clods under the drive wheel on rough, hand-tilled seedbeds will change the depth.

■ Assumptions

All assumptions are listed in SDL-CAL-001 section 1. The main ones are a 300 mm drive wheel over the lug tips (942.5 mm travel per turn), a 0.30 m/s cell fill limit, a Brixius rolling resistance model with cone indexes of 100 to 300 kPa, and the TRL 2 miss and double rates of 5 % and 4 %.